In fs breakdown, a thermodynamic temperature and state are established only after the laser pulse, when the thermalization of CB electron energy is completed. However, since hydrodynamic phenomena such as shock wave emission and bubble formation set in on a time scale of tens to about 100 ps, when sound propagation and particle movement become relevant compared to the source size, use of the thermodynamic temperature for characterizing the plasma with regard to hydrodynamic phenomena seems justified also for fs breakdown.

With ultrashort laser pulses, energy deposition is isochoric and no significant plasma expansion occurs during the laser pulse. Therefore, the mass density at the end of the laser pulse is known and U_{plasma} data can be used to determine T_{plasma} and p_{plasma}. Since thermal expansion takes place at sound velocity, the pulse duration τ_{L} must be shorter than the acoustic transit time through the plasma volume to fulfill this condition [188]. Therefore, it is often also denoted as 'stress confinement' condition. We will see in section IV.D that stress confinement applies not only for fs breakdown but also for luminescent ns plasmas.

Accepted Manuscript

3. Experimental methods

3.1. Experimental setup

The setup for the investigation of optical breakdown in water is presented in **Fig. 2**. Laser pulses of different durations, wavelengths, and temporal pulse shapes were focused through microscope objectives of various *NA*s into a cuvette containing double-distilled water (Braun, aqua ad iniectabilia) that had been filtered with a 0.22 µm microfilter (Millipore). Water-immersion objectives (Leica HCX APO L U-V-I) built into the wall of the cuvette provided diffraction-limited focusing conditions for wavelengths between 340 nm and 1100 nm. We investigated three *NA*s: *NA* = 0.8 (40x objective), *NA* = 0.5 (20x objective), and *NA* = 0.3 (10x objective). The use of large *NA*s avoided nonlinear beam propagation effects for ultrashort pulse durations [3,117,140] and stimulated Brillouin scattering for ns pulses, respectively [231-233].

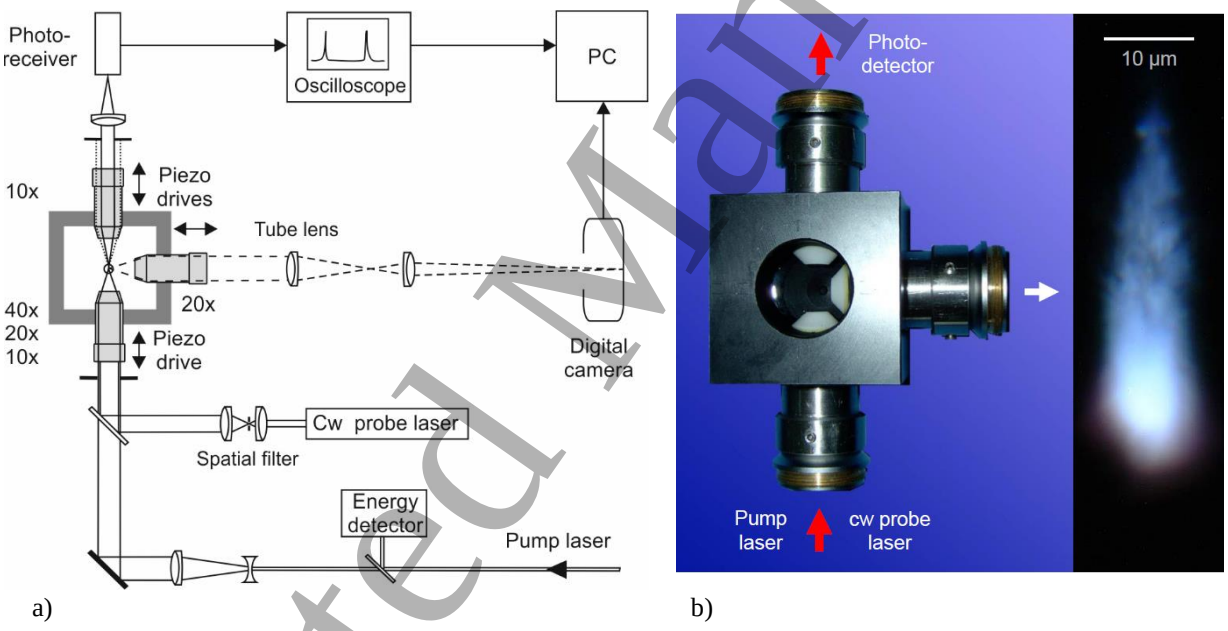

Figure 2 a) Experimental setup for investigating the energy dependence of femtosecond to nanosecond laser-induced breakdown in water at various pulse durations, wavelengths, *NA*s, and laser operation modes. b) Photograph of the water cell with inbuilt water immersion objective microscope objectives, and an example of a high-resolution plasma photograph taken through the 20x, *NA* = 0.5.

Orthogonal confocal adjustment of the objectives with long working distance (3.3 mm, 3.5 mm, and 3.6 mm, respectively, for the 40x, 20x, and 10x objectives) enabled high-resolution plasma photography in side view. To achieve confocality, the cylindrical ceramic cladding behind the conical front part was removed and replaced by a thinner stainless steel cover. Openings in the water cell made of

Accepted Manuscript

polyoxymethylene (POM) were exactly fitted to the diameter of the objectives to be watertight without further sealing. The *z*- position of the objectives could be controlled by piezo-driven devices.

The pump laser beam producing the breakdown events was expanded to overfill the rear entrance pupil of the focusing objective. This created an approximately uniform irradiance distribution corresponding to an Airy pattern in the focal plane. Single pulses were selected from the laser pulse train using a mechanical shutter. The pulse energy was adjusted by a combination of rotatable λ/2 plate and thin film polarizer and measured by diverting part of the incident laser pulse onto a calibrated energy meter (Ophir PD10-pJ, or Ophir PD 10).

The laser systems used in the experiments, together with the respective values of pulse duration, wavelength, and approximate values of the beam quality parameter M^2 are listed in Table 1.

Laser	Pulse duration (FWHM)	Wavelength (nm)	Beam quality parameter M^2
High Q Laser Production Yb:glass IC-1045-30-fs	280 fs	347	1.3
	306 fs	520	
	350 fs	1040	
TEEM Photonics PNV-001525-140	0.56 ns	355	1.2
CryLas FTSS 355-50 with IR option, and FTSS 355/532-50	0.93 ns	355	1.4
	0.95 ns	532	
	1.01 ns	1064	
Spectra Physics Nd:YAG Quanta Ray (instable resonator)	6.8 ns	355	1.6
	8.8 ns	532	
	11.2 ns	1064	

Table 1 Features of laser systems used in the experiments.

The Quanta Ray nanosecond Nd:YAG laser could either be run in multi-longitudinal-mode (mlm) operation or produce seeded, single-longitudinal-mode (slm) nanosecond pulses. The Teem Photonics and CryLas lasers emit amplified microchip ns laser pulses that operate intrinsically in single longitudinal mode due to the short resonator length. Occasional mode hops due to thermal drift of resonator length slightly shift the laser frequency but do not affect the single-mode emission itself. The High-Q Yb:glass fs laser system consists of a mode-locked (ml) oscillator and a 1-kHz regenerative amplifier. Mode-locked and slm emission go along with smooth temporal pulse shapes, while mlm emission exhibits

Accepted Manuscript